

- 1 A cliff train cabin is used to carry passengers up a slope as shown in Fig.1.1.

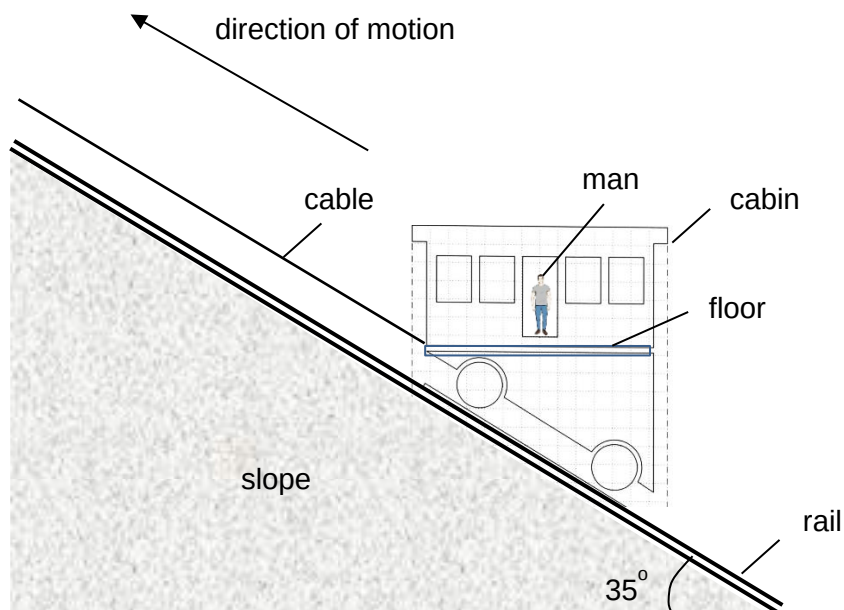


Fig. 1.1

The cable attached to the cabin pulls the cabin up the slope along the rail line which is inclined at 35° to the horizontal.

Initially, the cabin starts from rest and accelerates at 1.0 m s^{-2} for a time of 3.0 s . The cabin then moves at constant speed of 3.0 m s^{-1} for 100 s . Finally, the cabin decelerates to rest in 3.0 s .

The floor of the cabin is horizontal all the times. A man of mass 95 kg is standing upright on the floor of the cabin.

- (a) Calculate the vertical height moved by the man during the initial acceleration of the cabin.

height = m [3]

Distance moved **along the rail line**,

$$s = (0)(3.0) + \frac{1}{2}(1.0)(3.0)^2 = 4.5 \text{ m}$$

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Vertical height moved,

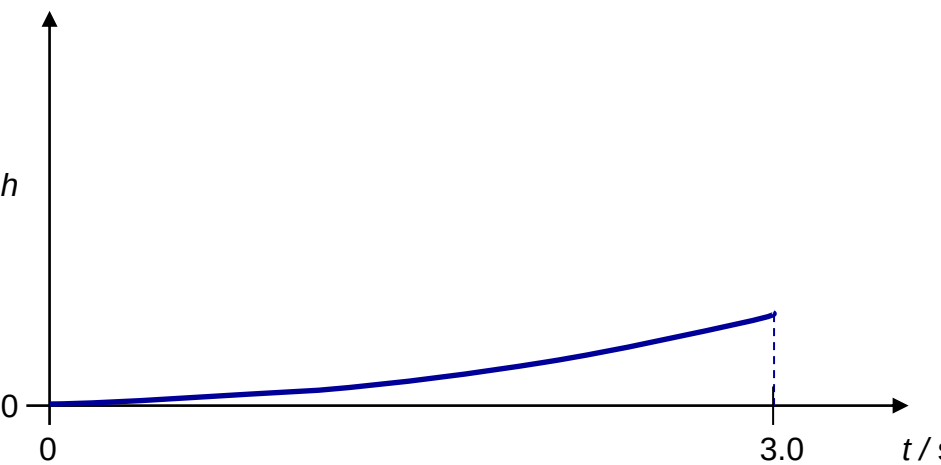
$$h = s \sin(35^\circ)$$

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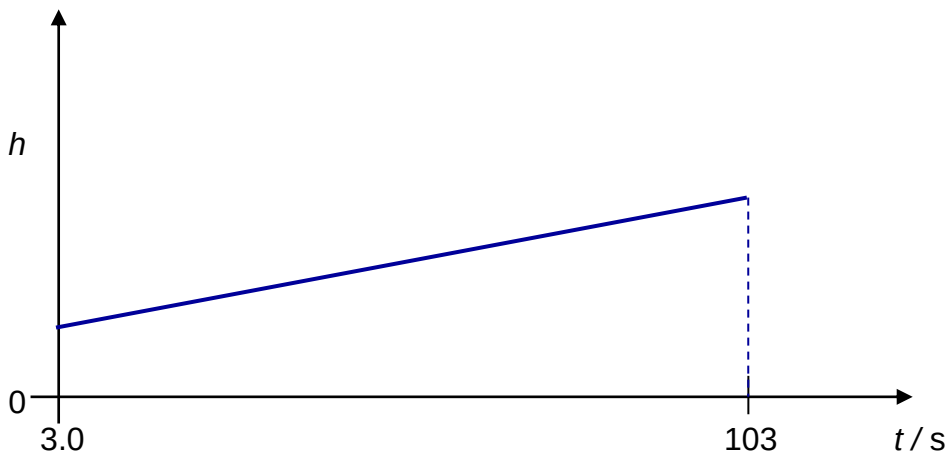
$$h = (4.5)(\sin 35^\circ) = 2.58109 = \mathbf{2.6 \text{ m}}$$

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	(b)	(i)	Calculate the normal reaction force acting on the man from the floor of the cabin when the cabin is moving at constant speed.	
			normal reaction = N	[1]
			$N = W = mg = 95 \times 9.81 = 931.95 = 930 \text{ N}$	1
		(ii)	Explain your working in (i).	
				
				[1]
			At constant speed, there is no acceleration, hence there is <u>no net</u> force acting on the <u>man</u>. There are two forces acting on the man: his weight and the normal reaction force of the floor on the man. Since the weight acts vertically down, the normal reaction force must be equal in magnitude and opposite in direction to the weight to produce a zero net force.	1
	(c)		Forces act on the man by the floor of the cabin.	
		(i)	State the forces for the man as the cabin accelerates.	
				[1]
			Frictional force and normal contact force by the floor, and, man's weight.	1
		(ii)	Explain how these forces produce the acceleration of the man.	
				
				[2]
			The vector sum of the frictional force, the normal contact force and the man's weight is non-zero and acts along the direction of the motion of the cabin. OR Normal contact force is greater than man's weight, causing a resultant force vertically upwards. The resultant of the horizontal leftwards acting frictional force and the resultant vertical force acts along the direction of the motion of the cabin to produce acceleration.	1 1 1 1

		(d)	The vertical height h of the man varies with time t . On the axes below, show qualitatively the variation with time t of height h for the motion of the man during	
		(i)	the acceleration,	
				[1]
		Solution:		
				1
		(ii)	the constant speed,	
				[1]

Solution:



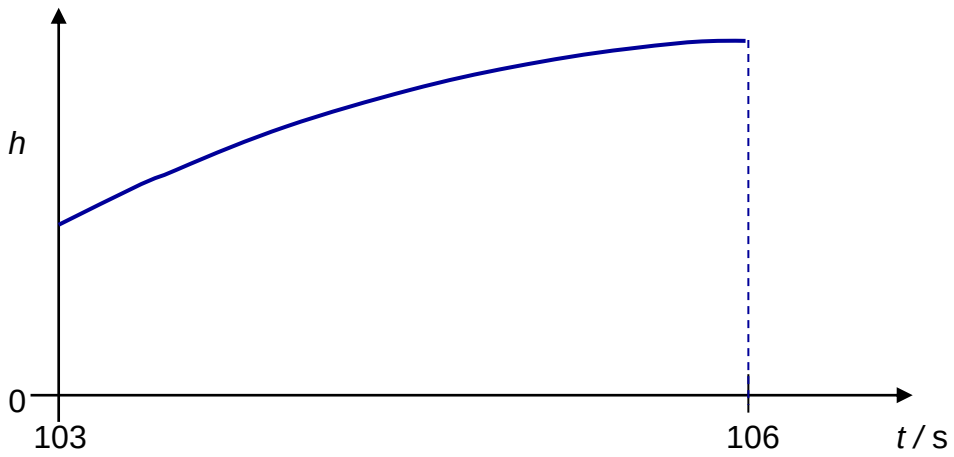
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(iii) the deceleration.



[1]

Solution:



1

2 A stationary electron is in a uniform field of force